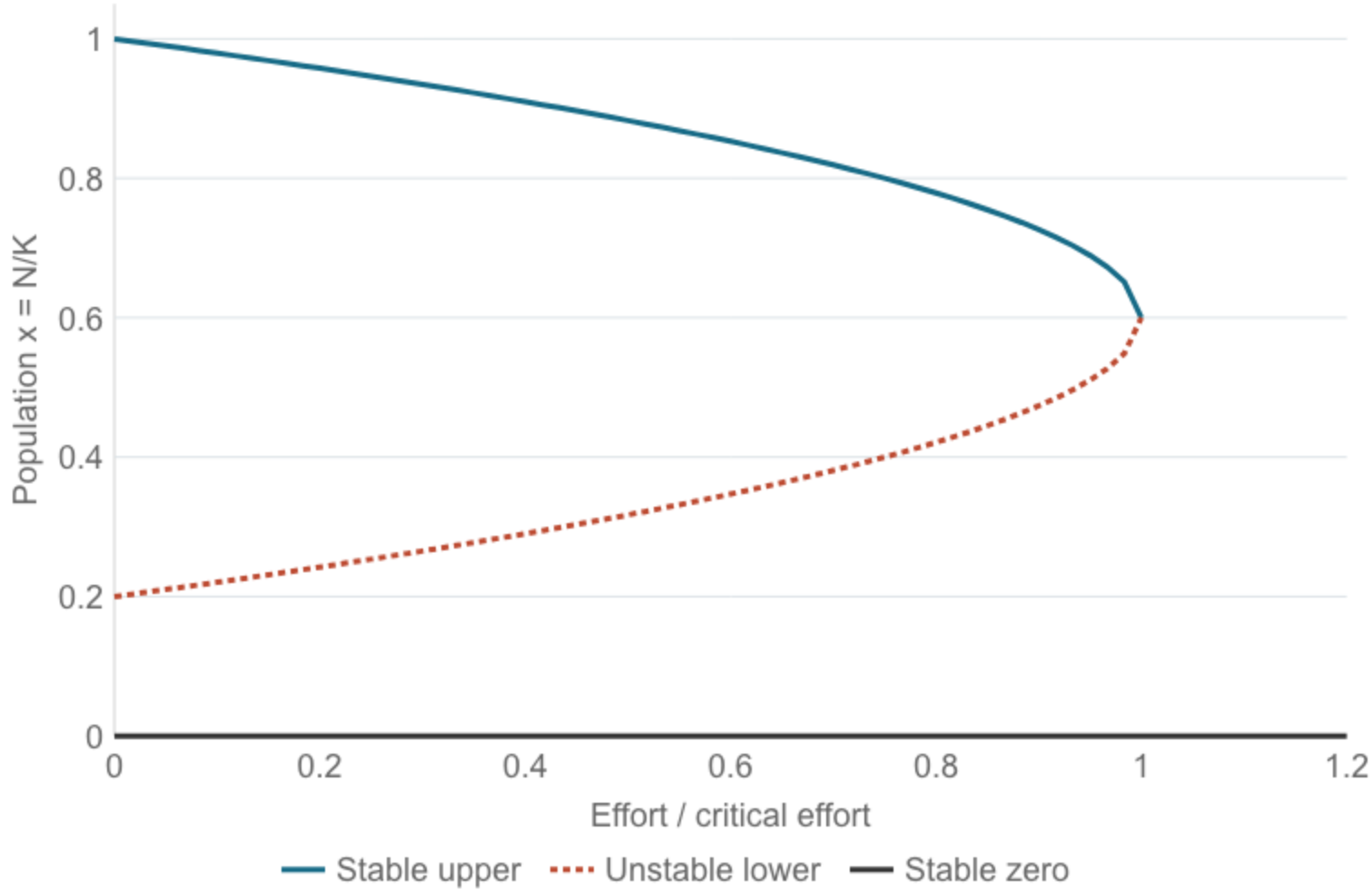
Model B: survival has a population threshold

Catch = hN

At low population, catch falls.
Natural growth can still be negative.

Lower positive root: unstable
Upper positive root: stable
Zero stays stable at every effort.



Equal catch and equal threshold fraction differ

Equal initial catch: $H = hN_0$

A and B both start at 131.28 tonnes/year; later catches can differ.

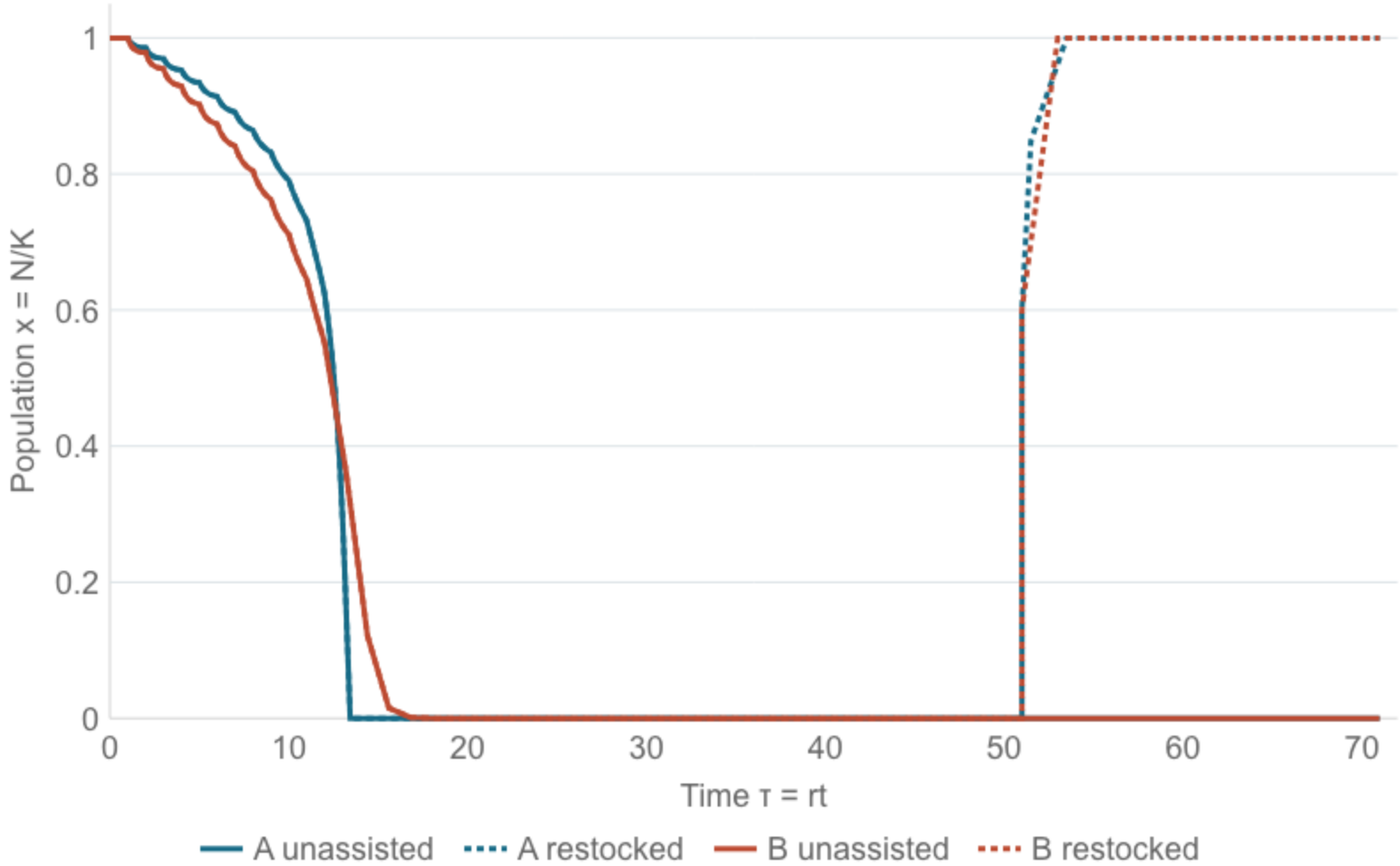
Half of each critical control

Model	Initial catch (t/year)	Final biomass (t)
A	131.28	920.90
B	180.00	882.84
Logistic	62.50	853.55

Different protocols answer different questions; neither gives a universal risk ranking.

Synthetic example · r = 0.5/year, K = 1000 tonnes, A = 200 tonnes, N_0 = 900 tonnes · horizon = 60 years

Recovery requires crossing the Allee threshold



Harvest rises to 1.3× critical, then returns to zero.

At τ = 51:
add 0.6K explicitly.
Recovery follows.

A: finite extinction.
B: asymptotic decline.
No spontaneous loop.

Synthetic RK45 histories · a = 0.2 · dwell = 1 · time τ = rt · detailed two-speed evidence in report